



Elastic Strings and Springs · Mark Scheme

Further Mechanics 1 Chapter 3: Elastic Strings and Springs. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Hooke's law and equilibrium

- 1 $T = \frac{20 \times 0.2}{0.8} \text{ M1}; = 5 \text{ N A1.}$
- 2 $T = mg = 2(9.8) \text{ M1}; 19.6 = \frac{\lambda \times 0.3}{1.2} \text{ A1}; \lambda = 78.4 \text{ N A1.}$
- 3 Thrust $= 1.5(9.8) = \frac{49x}{0.6} \text{ M1 A1}; x = 0.18$, so the length is 0.42 m A1.
- 4 $mg = \frac{0.1\lambda}{0.5} \text{ M1}; (m + 0.4)g = \frac{0.15\lambda}{0.5} \text{ M1};$ subtracting, $0.4g = 0.1\lambda$, so $\lambda = 39.2 \text{ A1}; m = 0.8 \text{ A1.}$
- 5(a) Let $AP = x$: $T_A = \frac{20(x-1)}{1} \text{ M1}; T_B = \frac{30(1.8-x)}{1.2} \text{ M1}; 20(x-1) = 25(1.8-x) \text{ M1}; x = \frac{13}{9} = 1.44 \text{ m A1.}$
- 5(b) $T_A = 20\left(\frac{4}{9}\right) \text{ M1};$ both tensions $= 8.89 \text{ N A1.}$
- 5(c) $T_A = 20(0.5) = 10 \text{ N, } T_B = 25(0.3) = 7.5 \text{ N M1};$ resultant $2.5 \text{ N towards A A1.}$

Section B – Elastic potential energy

- 6 $\text{EPE} = \frac{30(0.5^2)}{2(1.5)} \text{ M1}; = 2.5 \text{ J A1.}$
- 7 $\frac{40(0.6^2)}{2(0.8)} - \frac{40(0.2^2)}{2(0.8)} \text{ M1}; = 9 - 1 \text{ A1}; = 8 \text{ J A1.}$
- 8 $\text{EPE} = \frac{20(0.1^2)}{2(0.4)} = 0.25 \text{ J M1}; \frac{1}{2}(0.5)v^2 = 0.25 \text{ A1}; v = 1 \text{ ms}^{-1} \text{ A1.}$
- 9 Let the greatest extension be x : loss in GPE = gain in EPE $\text{M1}; 0.5(9.8)(1+x) \text{ A1} = \frac{19.6x^2}{2(1)} \text{ A1}; 2x^2 - x - 1 = 0$, $x = 1$, so greatest distance 2 m A1.
- 10(a) $70(9.8)(20+x) = \frac{1680x^2}{2(20)} \text{ M1 A1}; 42x^2 - 686x - 13720 = 0$, i.e. $3x^2 - 49x - 980 = 0 \text{ M1}; x = 28$, so 48 m below O A1.
- 10(b) $\frac{1}{2}(70)v^2 = 70(9.8)(30) - \frac{1680(10^2)}{40} \text{ M1}; v^2 = 468, v = 21.6 \text{ ms}^{-1} \text{ A1.}$
- 10(c) Maximum speed when $T = mg$: $\frac{1680e}{20} = 686 \text{ M1}; e = 8.17 \text{ m A1}; \frac{1}{2}(70)v^2 = 686(28.17) - \frac{1680(8.167^2)}{40} \text{ M1}; v = 21.7 \text{ ms}^{-1} \text{ A1.}$

Roar-level challenge

- 11 Work against friction per metre $= 0.25 \times 2(9.8) \cos 30^\circ = 4.24 \text{ N B1};$ with greatest extension x : $2(9.8)(1.5+x) \sin 30^\circ = 4.24(1.5+x) + \frac{36x^2}{2(1.5)} \text{ M1 B1}; 12x^2 - 5.556x - 8.335 = 0 \text{ M1}; x = 1.10 \text{ m A1.}$